SPECTRAL GARDEN

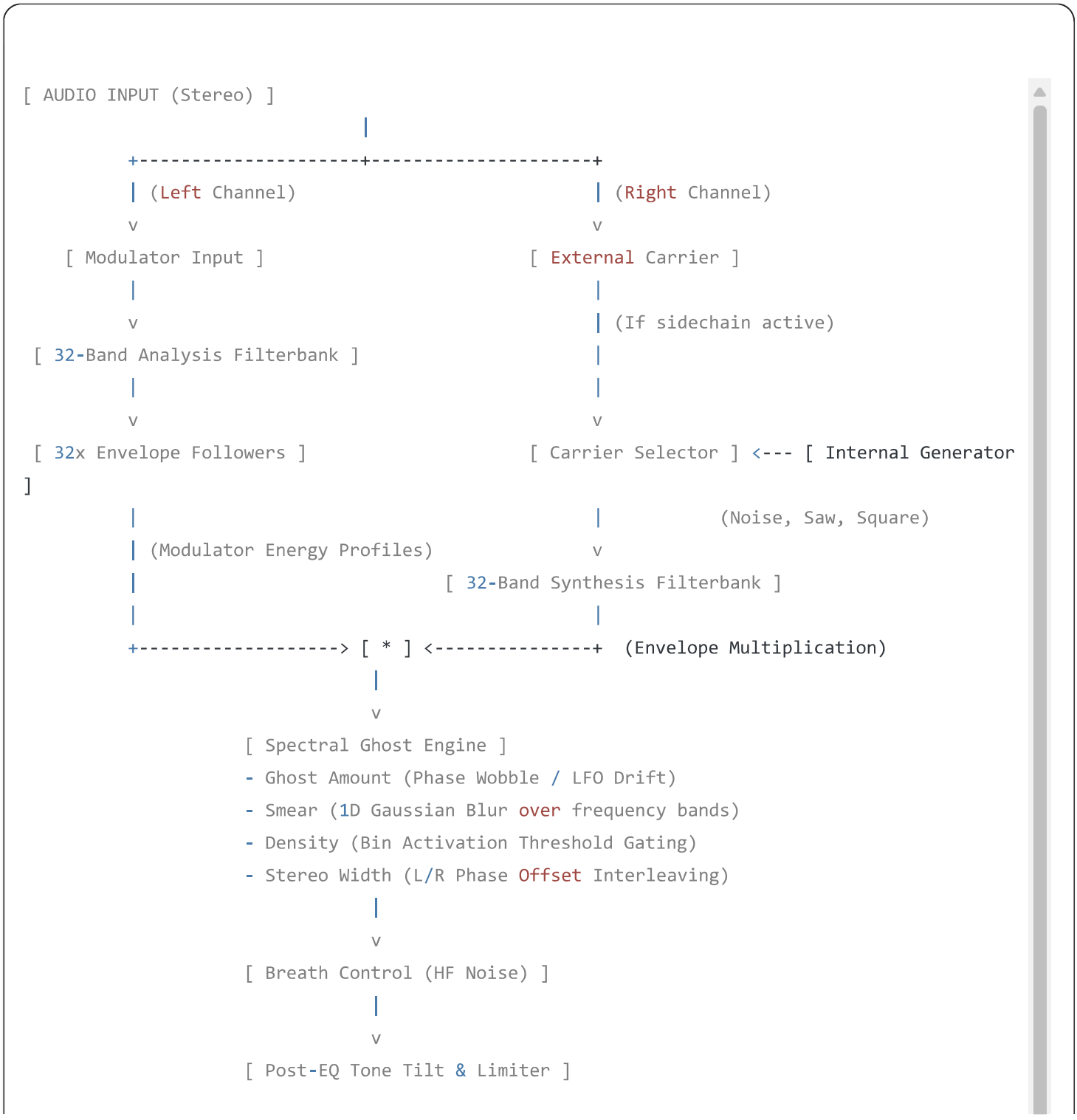
# Wraithgate Vocoder (v1.0.0)

*A 32-Band Spectral, Ghost-Driven Vocoder for JUCE 8.14.0 and Visual Studio 2026*

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## 1. System Architecture & Signal Flow

The **Wraithgate Vocoder** operates as a real-time spectral morphing processor, crossing the frequency characteristics of an incoming **Modulator** (traditionally a vocal/speech source) with a **Carrier** (internal synthesized waves or an external sidechained instrument) through a highly optimized 32-band filterbank.



```
                    |
                    v
          [ Dry/Wet Mix Crossfade ]
                    |
                    v
            [ OUTPUT BUFFER ]
```

### Analysis Filterbank & Envelope Detection

The Modulator signal is processed through a bank of 32 bandpass filters spaced logarithmically from

$$80\ \text{Hz}$$

to $12\ \text{kHz}$. Each band isolates a specific frequency region, feeding its filtered sub-band into a dedicated **Envelope Follower**. The envelope follower implements asymmetrical attack and release time constants to track transient rises and smooth decays without inducing high-frequency chatter or aliasing.

### Carrier Generation & Synthesis Filterbank

The Carrier signal is either synthesized internally via the low-latency multi-mode oscillator (Noise, Sawtooth, Square) or accepted externally (by routing the right input channel as a sidechain). This Carrier is processed through a twin bank of 32 synthesis bandpass filters. By default, these filter frequencies match the analysis bands, but they can be offset dynamically using the **Formant Shift** control.

### Envelope Multiplication

The output of each synthesis band is multiplied by the corresponding envelope extracted from the modulator. This step impresses the time-varying spectral contours of the modulator directly onto the carrier.

### Spectral Ghost Engine

Once multiplied, the bands pass through the **Ghost Engine**, which modulates the spectral signal in the frequency domain:

- **Phase Drift (LFO)**: Random-walking, slow-moving LFOs apply micro-frequency offsets per band, mimicking physical and phase-coherent drift.
- **Gaussian Smear**: A 1D blur kernel disperses envelope energy from dominant bands into adjacent frequency channels, creating a haunting, diffuse wash.
- **Spectral Density**: Performs stochastic gating on individual bands to simulate intermittent spectral dropouts and fading bin activations.
- **Ghost Imaging**: Applies interleaved L/R phase offsets to widen the stereo field dynamically.

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## 2. Control Parameter Reference

### Modulator & Global Controls

- **Input Gain (-24.0 dB to +24.0 dB)**: Controls the incoming amplitude of the modulator signal before it hits the analysis filterbank.

- **Output Gain (-24.0 dB to +24.0 dB)**: Post-limiter gain stage adjusting the final output level.
- **Mix (0.0 to 1.0)**: Blends the unprocessed dry signal (modulator input) with the processed vocoded signal.
- **Tone (0.0 to 1.0)**: A post-EQ tilt filter. Setting this to $0.5$ is flat. Below $0.5$ boosts low frequencies while attenuating high frequencies; above $0.5$ tilts the spectrum toward bright, airy spectral highs.
- **Bypass (On/Off)**: Initiates a $50\ \text{ms}$ click-free linear crossfade directly between the dry input buffer and wet output buffer.
- **Oversampling (Off, 2x, 4x, 8x)**: Processes the entire DSP path at an elevated sample rate using a polyphase IIR half-band filterbank. This dramatically reduces filter-aliasing artifacts when modulating sharp waveforms.
- **Eco Mode (On/Off)**: Decimates the envelope follower and ghost engine calculations to every second sample. Useful for conserving processing threads on older CPU architectures.

## Carrier Controls

- **Carrier Source (Noise, Saw, Square, External)**: Selects the sound source fed into the synthesis filterbank. "External" routes the right stereo input channel as the carrier.
- **Carrier Pitch (40.0 Hz to 440.0 Hz)**: Adjusts the fundamental frequency of the internal Saw and Square oscillators.
- **Formant Shift (-12.0 st to +12.0 st)**: Transposes the center frequencies of the synthesis filterbank relative to the analysis filterbank. Shifting upward yields brighter, childlike speech timbres; shifting downward creates deep, guttural tones.
- **Breath Control (0.0 to 1.0)**: Injects high-frequency white noise into the upper bands of the synthesis output when high-frequency energy is detected in the modulator, ensuring sibilant consonants ("s", "t", "ch") remain highly intelligible.

## Ghost Engine Controls

- **Ghost Amount (0.0 to 1.0)**: Increases the LFO-driven phase wobble, causing bandpass frequencies to drift independently, generating detuned, chorus-like dispersion.
- **Smear (0.0 to 1.0)**: Adjusts the radius of the 1D frequency-domain Gaussian blur. Higher settings smooth out sharp formant transitions into an ethereal, pad-like blur.
- **Density (0.0 to 1.0)**: Controls the probability threshold of bin activation. Lower values force random bands to drop out, creating a flickering, disintegrating spectral effect.
- **Width (0.0 to 1.0)**: Introduces LFO phase offsets between the left and right synthesis filters, generating wide stereo widening.
- **Attack Time (1.0 ms to 1000.0 ms)**: Configures the rise time of the envelope followers. Faster times capture transient plosives; slower times lag, creating soft, swelling transitions.

- **Release Time (5.0 ms to 3000.0 ms)**: Configures the decay rate of the envelope followers. High values result in long, atmospheric tails.

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# 3. Optimizations & Performance Tuning

To keep CPU load minimal (~1% on typical platforms) while rendering high-fidelity animations, the plugin separates DSP calculations by rate:

```
[ REAL-TIME HOST PROCESS BLOCK ]
  |
  +---> [ Slow Block-Rate (e.g., every 64-256 samples) ]
  |       - Read parameters & update LFOs
  |       - Re-calculate IIR filter coefficients (std::sin, std::cos, std::pow)
  |       - Re-map analysis & synthesis filter target frequencies
  |
  +---> [ Fast Sample-Rate (per sample processing) ]
          - Run simple 1-pole Envelope Followers
          - Run lightweight IIR processSample() loops
          - Perform multiply-accumulate operations on envelopes
          - Apply dry/wet crossfades
```

## Block-Rate Coefficient Calculations

Calculating IIR coefficients using standard trigonometric methods is an expensive operation. To solve this, the **Wraithgate Vocoder** updates its filter coefficient pointers **once per block** rather than per sample. Since formant transpositions and LFO drifts occur on sub-audio timescales, block-rate updates are highly optimized and acoustically seamless.

## GPU Accelerated UI Compositing

The user interface utilizes a `juce::OpenGLContext` attached to the top-level editor. Rather than instantiating child context views that compete for window layering handles, the entire 2D component tree is drawn directly to a GPU frame buffer. Standard vector drawings (like the central occult sigil, ghost particle systems, and the 32 spectral bands) are rendered via GPU-accelerated pipelines, bypassing CPU-based software rendering entirely.

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# 4. Compilation & Integration Guide

## Required Software IDE & Toolsets

- **JUCE Framework**: version 8.14.0 or newer.
- **IDE**: Microsoft Visual Studio 2026.
- **C++ Standard**: C++20.

## Required JUCE Modules

To successfully compile, verify that the following modules are added and enabled inside your `.jucer` project:

- `juce_core`
- `juce_data_structures`
- `juce_events`
- `juce_graphics`
- `juce_gui_basics`
- `juce_gui_extra`
- `juce_dsp`
- `juce_audio_basics`
- `juce_audio_devices`
- `juce_audio_formats`
- `juce_audio_processors`
- `juce_audio_utils`
- `juce_opengl`

## MIDI CC Mapping Implementation

The plugin features an integrated MIDI Learn system.

1. **To Map**: Right-click any dial on the interface. A popup menu will materialize. Select "Learn MIDI CC". The plugin will automatically monitor incoming MIDI messages and map the first detected Controller Number (CC) to that target parameter.
2. **To Unmap**: Right-click a mapped dial and select "Forget MIDI CC" to clear the assignment.
3. **Storage**: MIDI assignments are automatically serialized into the plugin's host state XML chunk and restored when reloading a session.

## Preset Directories & File Structures

User presets are written as raw JSON files to prevent platform dependencies.

- **Windows File Path**: `C:\Users\<Username>\Documents\Coven Audio\Wraithgate Vocoder\Presets\`
- **Format**: Standard UTF-8 JSON.
- **Action**: To share presets, copy `.json` files directly into this directory and refresh the instance or click any category inside the preset browser to update the listing.